

server addresses, head over to the network adapter properties. Double-click on 'Internet Protocol Version 4 (TCP/IPv4)' and a window will pop up. Click on the radio button 'Use the following DNS server addresses' and now enter the previously mentioned addresses. If you wish to add more

that can be done by clicking in Advanced and going to the DNS tab. Here you can add as many DNS servers as you want and sort them according to a connection priority. This can be used in case any of the servers face downtime then some server will continue running. Do note, sometimes the router might override the DNS server, so it would be useless to implement on the PC.

IMPROVE THROUGHPUT USING JUMBO FRAMES

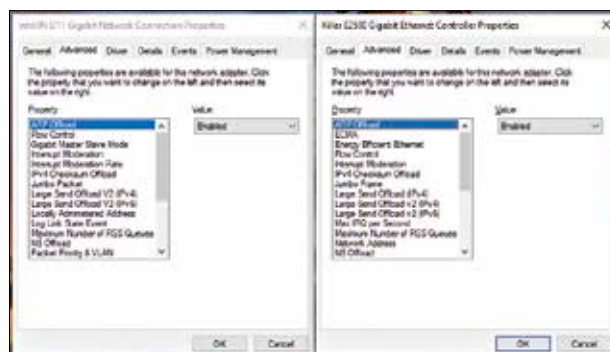
This is where we get into the advanced techniques to optimise your network. When you open the Properties of the network adapter, there's a 'Configure' button on the top. On clicking, it will take you to a new window with more functions. Head over to the 'Advanced' tab and you'll be listed with a plethora of



Obtain higher throughput with Jumbo Frame

functions. These won't be the same for everyone since it's dependent on the network adapter. We'll be taking a look at the most important ones.

Now, click on Jumbo Frame from the list. Enabling Jumbo Frames based on the desired maximum transmission unit or MTU from the dropdown list on the right will make your network more efficient and increase the throughput. It's supported only on local area networks that support at least 1Gbps. It's also dependent on your adapter vendor. The only downside to this is that all the devices in the network need to have jumbo frames enabled at the same MTU. Otherwise, it would lead to packet loss and in turn hamper your connection. However, you should enable them since it's quite possible that the devices already support it.



Different network adapters will support different functionalities

INCREASE TRAFFIC REGULATION EFFICIENCY WITH FLOW CONTROL

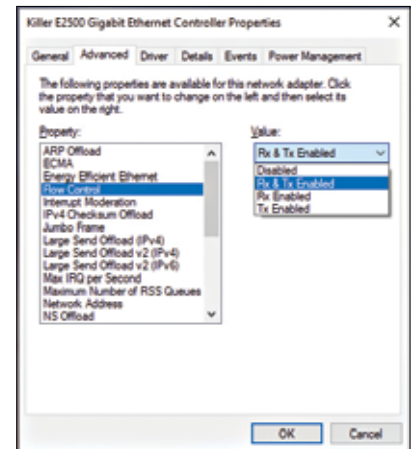
Disabling Flow Control might help with the increase in the efficiency of traffic regulation for connections. It's said that the implementation of Flow Control is buggy in certain network adapters and hence, it affects the connection. Disabling it should reduce timeouts and improve throughput. However, before disabling, you should read more about how well it's implemented in your system's network adapter.

ENABLE RECEIVE SIDE SCALING

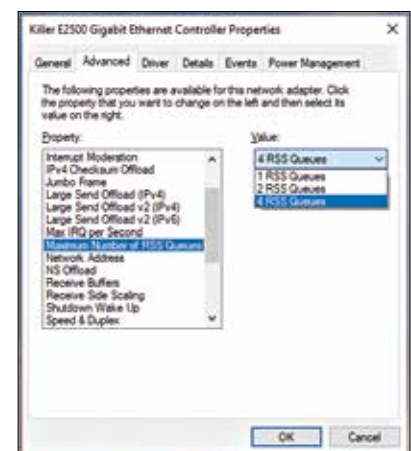
Side scaling allows your system to distribute all the receive data processing to multiple processors or processor cores. In order to do so, 'Receive Side Scaling' has to be enabled which usually is the default state since almost all systems have more than a single processor now.

By distributing the receive data, there's higher efficiency which leads to higher performance. If this is disabled, the burden of processing the receive data will fall on a single core and go on to affect system cache utilisation. Older CPUs usually faced an issue with RSS enabled, especially while playing games. But the CPUs nowadays are capable of handling it, so there's no reason to disable RSS.

Some adapters might support RSS queue settings. Here, you're allowed to change the maximum number of RSS queues as listed under 'Maximum Number of RSS Queues'. It would be best to choose two queues that will ensure good throughput and low CPU utilisation. This is where you can do some trial and error



Reduce timeouts and improve throughput



Selecting 2 RSS Queues are recommended

if more than two queues are supported. If the CPU utilisation is high, then you should stick to two queues.

SELECT SPEED AND DUPLEX OF YOUR CONNECTION

You can choose the speed of your network adapter along with the communication type. If your adapter supports Gigabit speeds, it will automatically appear in